

Data-Driven Turbulence Modeling Using Tensor-based Neural Network with Consideration of The Geometric Effect

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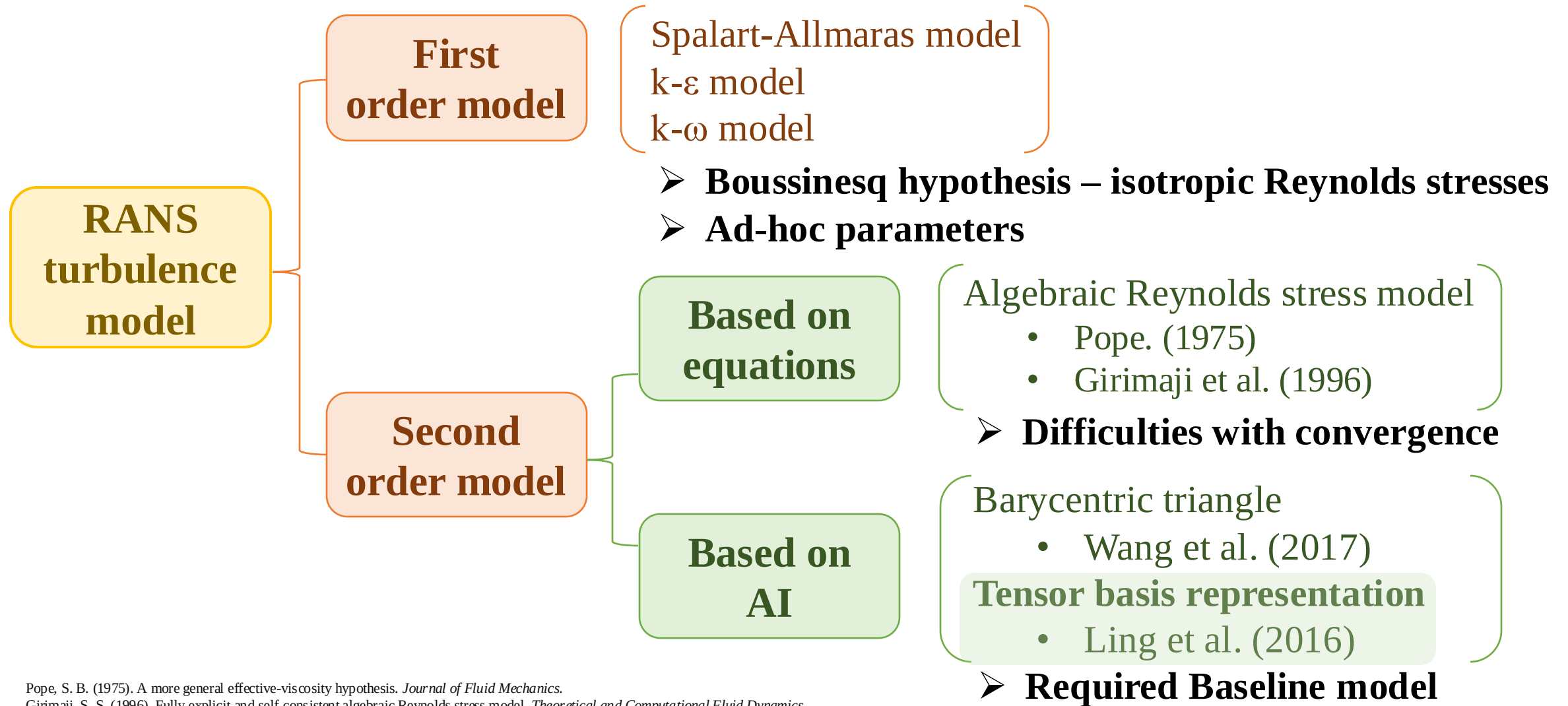


Outline

- 1. Introduction**
- 2. Methodology**
- 3. Result & Discussion**
- 4. Conclusion & Future work**
- 5. References**



Introduction - Literature



- Pope, S. B. (1975). A more general effective-viscosity hypothesis. *Journal of Fluid Mechanics*.
- Girimaji, S. S. (1996). Fully explicit and self-consistent algebraic Reynolds stress model. *Theoretical and Computational Fluid Dynamics*.
- Ling, J., Kurzawski, A., & Templeton, J. (2016). Reynolds averaged turbulence modelling using deep neural networks with embedded invariance. *Journal of Fluid Mechanics*.
- Wang, J.-X., Wu, J.-L., & Xiao, H. (2017). Physics-informed machine learning approach for reconstructing Reynolds stress modeling discrepancies based on DNS data. *Physical Review Fluids*.

Introduction - Motivation

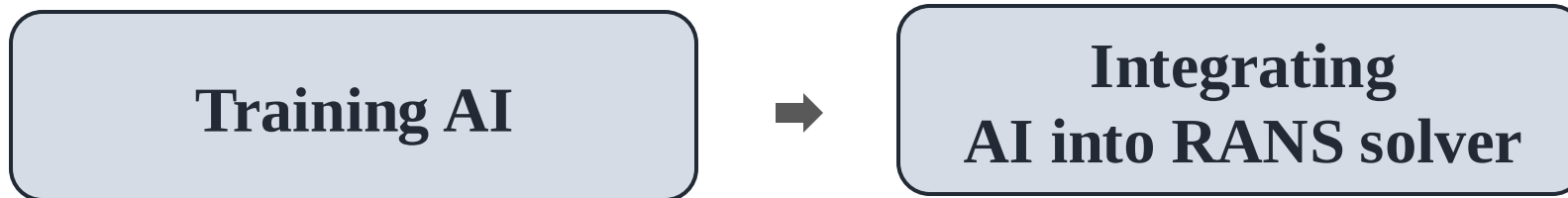
Past methods for tensor basis representation



- Computationally inefficient
- Ignores the inherent fitting capability of AI

Our method

- Using AI to predict anisotropic Reynolds stresses
- Integrating AI model into RANS solver



Methodology - Governing equation

➤ RANS equation:

$$\frac{\partial U_i^*}{\partial t^*} + U_k^* \frac{\partial U_i^*}{\partial x_k^*} = \frac{-1}{\rho} \frac{\partial P^*}{\partial x_i^*} + \frac{1}{Re} \frac{\partial^2 U_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial}{\partial x_j^*} \tau_{ij}^*$$

$$u_i^* = U_i^* + \langle u_i^* \rangle \quad \tau_{ij}^* = \overline{\langle u_i^* \rangle \langle u_j^* \rangle} \quad * : \text{Dimensionless}$$

➤ Reynolds stresses anisotropic tensor:

$$b_{ij} = \frac{\tau_{ij}}{2k} - \frac{1}{3} \delta_{ij} = \tau_{ij}^* - \frac{1}{3} \delta_{ij}$$

$$\tau_{ij}^* = b_{ij} + \frac{1}{3} \delta_{ij}$$

➤ Tensor basis representation, (Pope 1975):

$$b_{ij} = B_{ij}(S_{ij}^*, R_{ij}^*)$$

$$B_{ij}(S_{ij}^*, R_{ij}^*) = \sum_{k=1}^{10} G^{(k)}(\lambda_m | m = 1, 2, \dots, 5) T_{ij}^{*(k)}$$

S_{ij} : Strain-rate tensor

R_{ij} : Rotation-rate tensor

λ_m : Invariants

T_{ij} : Integrity tensor

➤ Final Form

$$\tau_{ij}^* = b_{ij} + \frac{1}{3} \delta_{ij} = b_{ij} \Rightarrow \tau_{ij}^* = \sum_{k=1}^{10} G^{(k)}(\lambda_m) T_{ij}^{*(k)}$$

$\frac{1}{3} \delta_{ij}$: Add to the pressure term

• Pope, S. B. (1975). A more general effective-viscosity hypothesis. *Journal of Fluid Mechanics*.

Methodology - Computational Process

➤ Selection of dimensionless parameters ($[u]$, $[l]$):

- No baseline model is required
- Preserving lower-order dominance
- $\tau_L = \frac{[l]}{[u]}$ is the large eddy time scale, $\tau_\eta = \frac{[k]}{[\varepsilon]}$ is the smaller eddy time scale defined as $\frac{\tau_\eta}{\tau_L} = Re^{\frac{1}{2}}$

$$\mathbf{b}_{ij} = \mathbf{B}_{ij,k\varepsilon}(\mathbf{S}_{ij,k\varepsilon}^*, \mathbf{R}_{ij,k\varepsilon}^*)$$

$$\mathbf{S}_{ij,k\varepsilon}^* = \frac{[k]}{[\varepsilon]} \mathbf{S}_{ij}, \quad \mathbf{R}_{ij,k\varepsilon}^* = \frac{[k]}{[\varepsilon]} \mathbf{R}_{ij}$$

$$\mathbf{b}_{ij} = \mathbf{B}_{ij,ul}(\mathbf{S}_{ij,ul}^*, \mathbf{R}_{ij,ul}^*)$$

$$\mathbf{S}_{ij,ul}^* = \frac{1}{Re^{\frac{1}{2}}} \frac{[l]}{[u]} \mathbf{S}_{ij}, \quad \mathbf{R}_{ij,ul}^* = \frac{1}{Re^{\frac{1}{2}}} \frac{[l]}{[u]} \mathbf{R}_{ij}$$

➤ Optimal eddy viscosity:

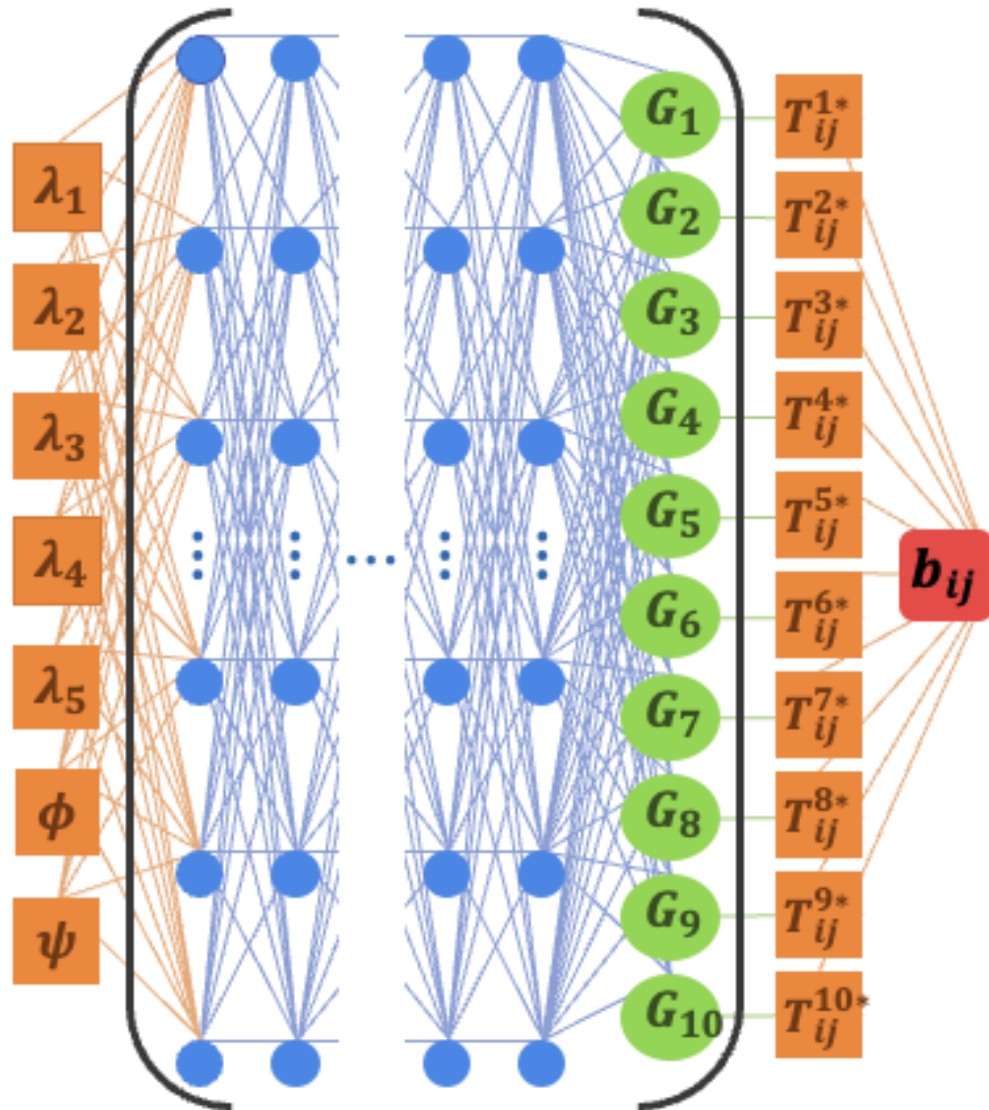
- Stable in the computational process

$$\mathbf{b}_{ij} = \sum_{k=1}^{10} \mathbf{G}^{(k)}(\lambda_m) \mathbf{T}_{ij}^{*(k)} \approx -2\mathbf{v}_t^* \mathbf{S}_{ij,ul}^*$$

$$\mathbf{v}_{t,opt}^* = \arg \min_{\mathbf{v}_t^* \geq 0} \|\mathbf{b}_{ij} - (-2\mathbf{v}_t^* \mathbf{S}_{ij,ul}^*)\| \Rightarrow -\frac{1}{2} \frac{\mathbf{b}_{ij} \cdot \mathbf{S}_{ij,ul}^*}{\mathbf{S}_{ij,ul}^* \cdot \mathbf{S}_{ij,ul}^*}$$

$$\boldsymbol{\tau}_{ij}^* = \mathbf{b}_{ij} = -2\mathbf{v}_{t,opt}^* \mathbf{S}_{ij,ul}^*$$

Methodology - AI Architecture



➤ Tensor Basis Neural Network (TBNN) Design:

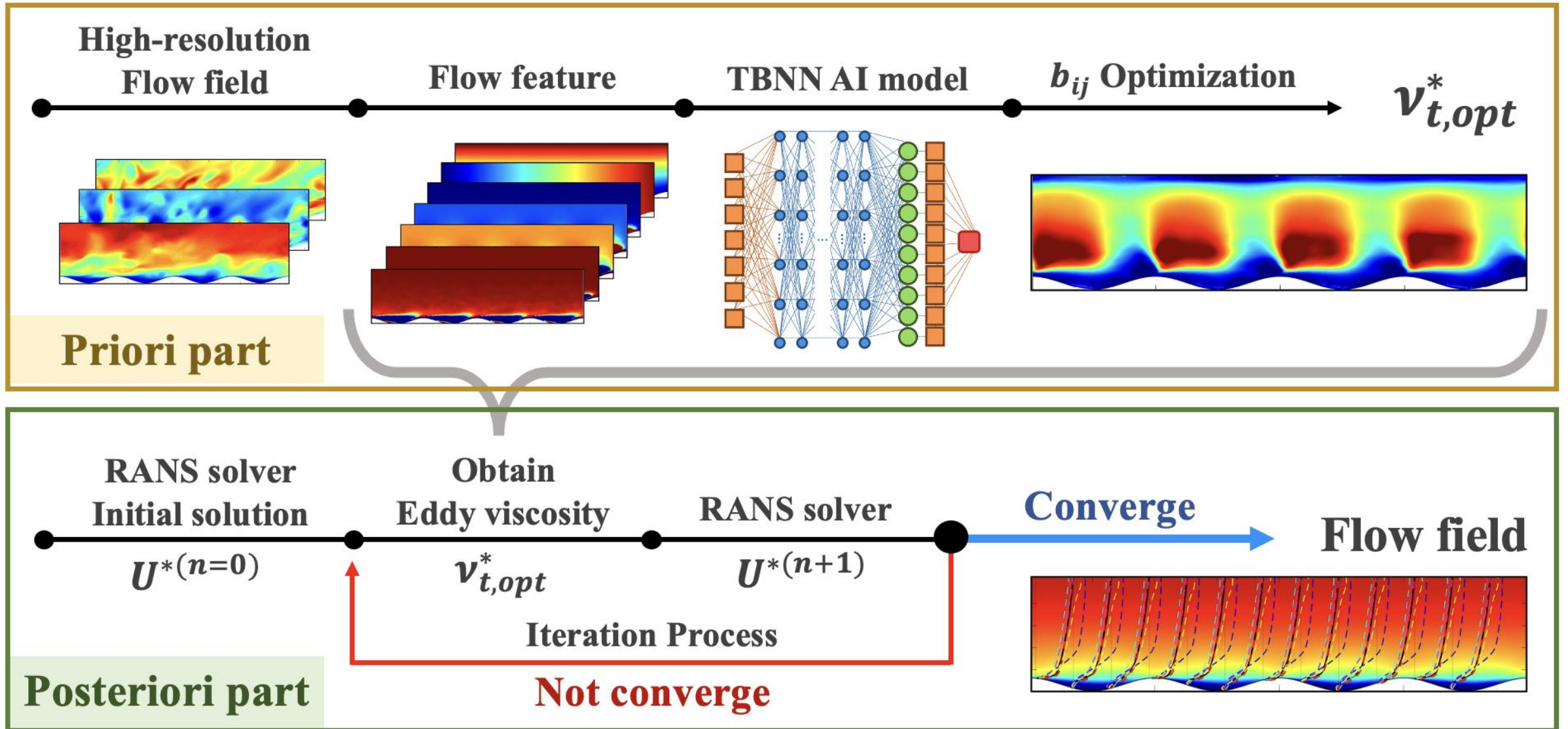
- Tensor basis representation:

$$b_{ij}(S, R) = \sum_{k=1}^{10} G^{(k)}(\lambda_m | m = 1, 2, \dots, 5) T_{ij}^{*(k)}$$

- Adding ϕ, ψ as Training feature

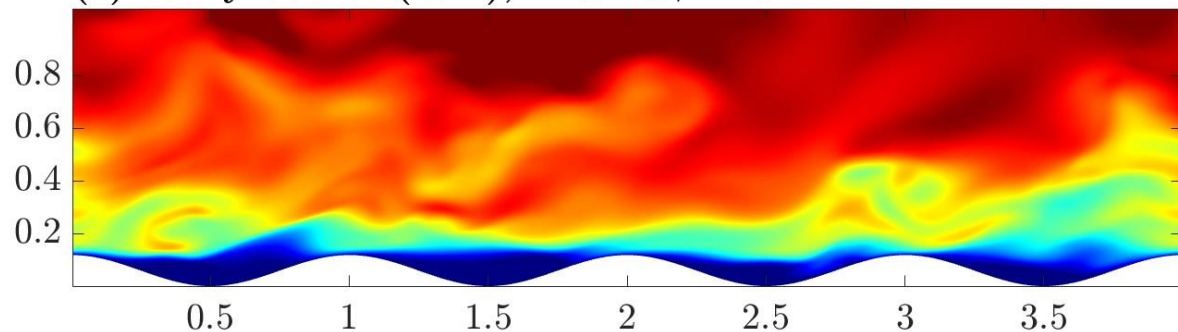
Invariants	Linearly Independent Tensor	
$\lambda_1 = \{S^2\}$	$T_{ij}^{1*} = S$	$T_{ij}^{6*} = R^2 S + SR^2 - \frac{2}{3} I\{SR^2\}$
$\lambda_2 = \{S^3\}$	$T_{ij}^{2*} = SR - RS$	$T_{ij}^{7*} = RSR^2 - R^2 SR$
$\lambda_3 = \{R^2\}$	$T_{ij}^{3*} = S^2 - \frac{1}{3} I\{S^2\}$	$T_{ij}^{8*} = SRS^2 - S^2 RS$
$\lambda_4 = \{R^2 S\}$	$T_{ij}^{4*} = R^2 - \frac{1}{3} I\{R^2\}$	$T_{ij}^{9*} = R^2 S^2 + S^2 R^2 - \frac{2}{3} I\{S^2 R^2\}$
$\lambda_5 = \{S^2 R^2\}$	$T_{ij}^{5*} = RS^2 - S^2 R$	$T_{ij}^{10*} = RS^2 R^2 - R^2 S^2 R$
ϕ, ψ		

Methodology - Framework

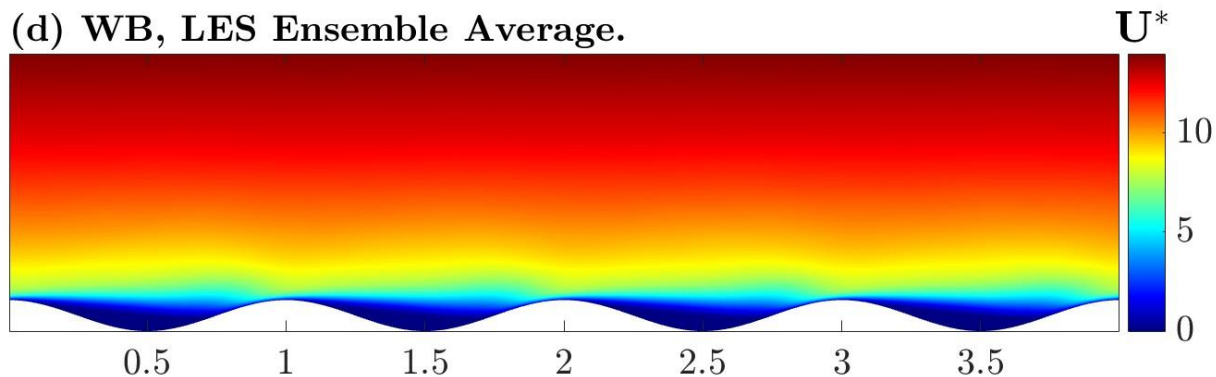


Result & Discussion - LES data

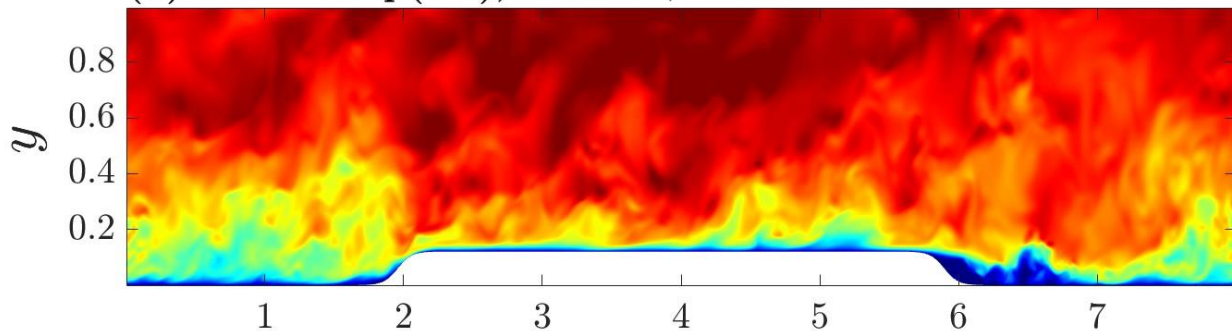
(a) Wavy bottom(WB), LES. $Re_\tau = 251$.



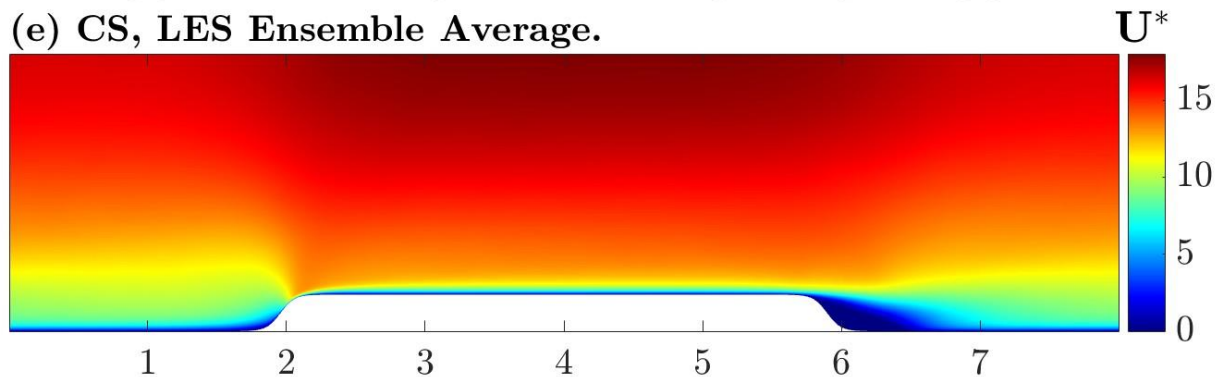
(d) WB, LES Ensemble Average.



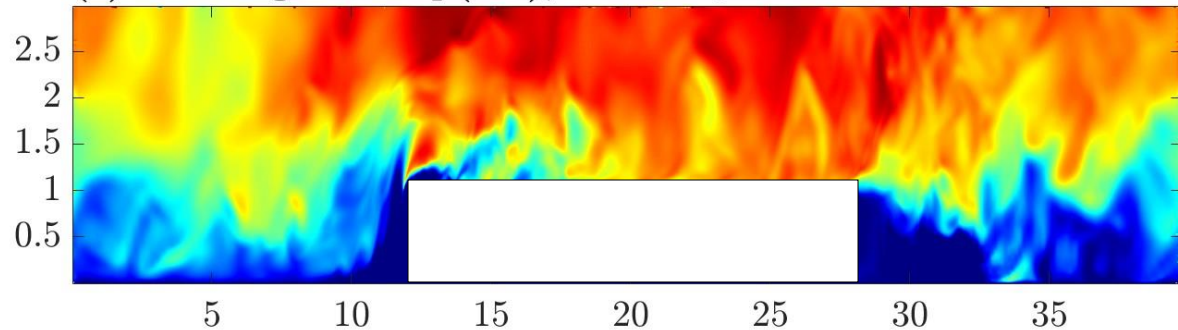
(b) Curve Step(CS), LES. $Re_\tau = 710$.



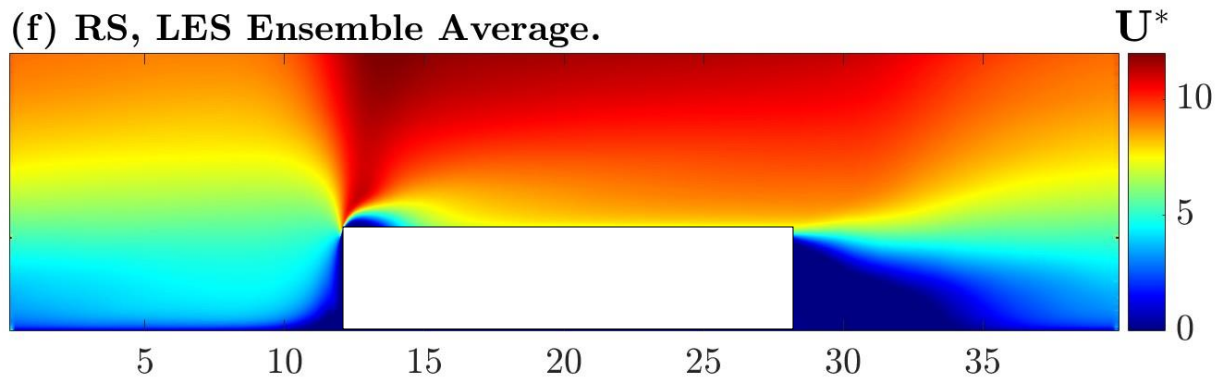
(e) CS, LES Ensemble Average.



(c) Rectangular step(RS), LES. $Re_\tau = 1325$.



(f) RS, LES Ensemble Average.

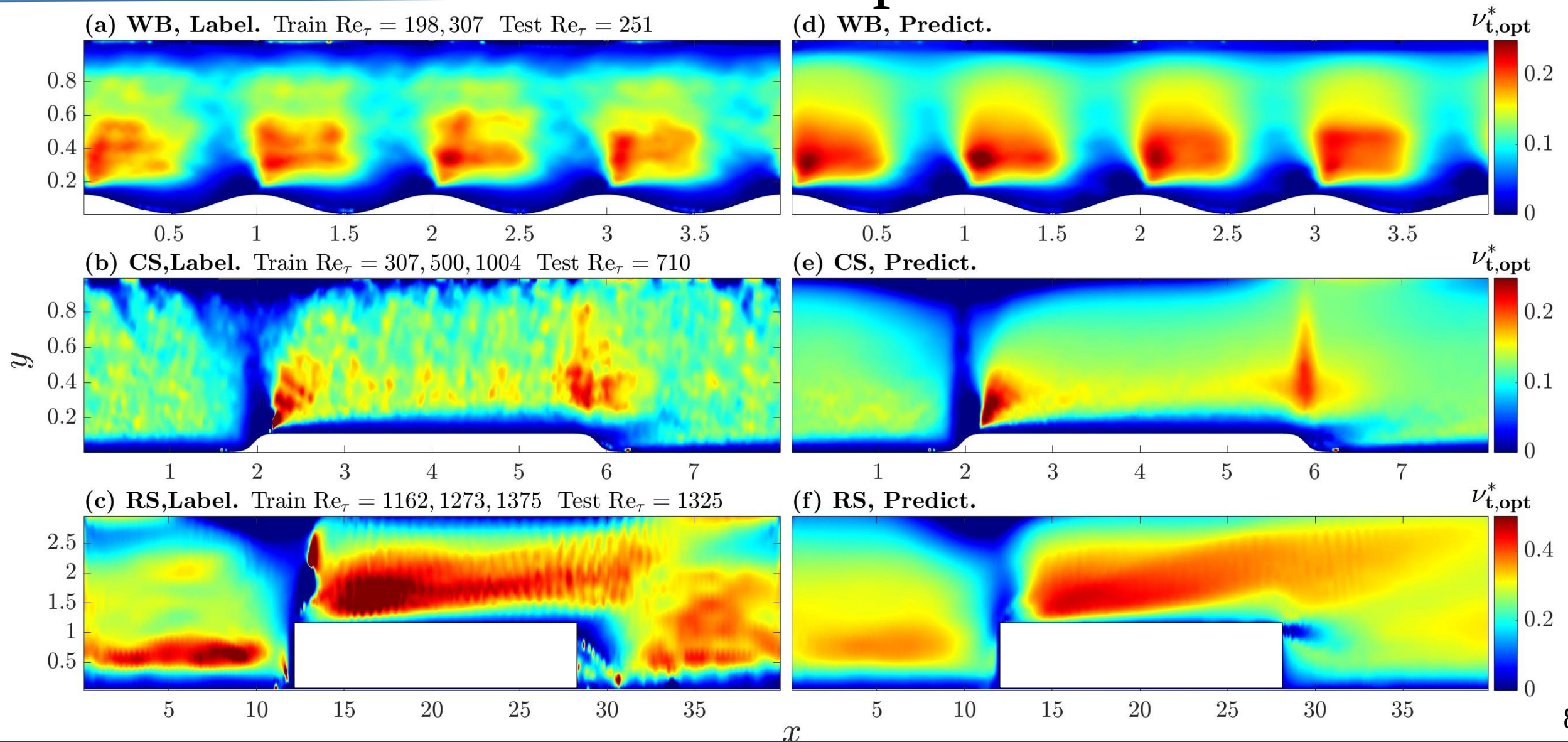


x

7

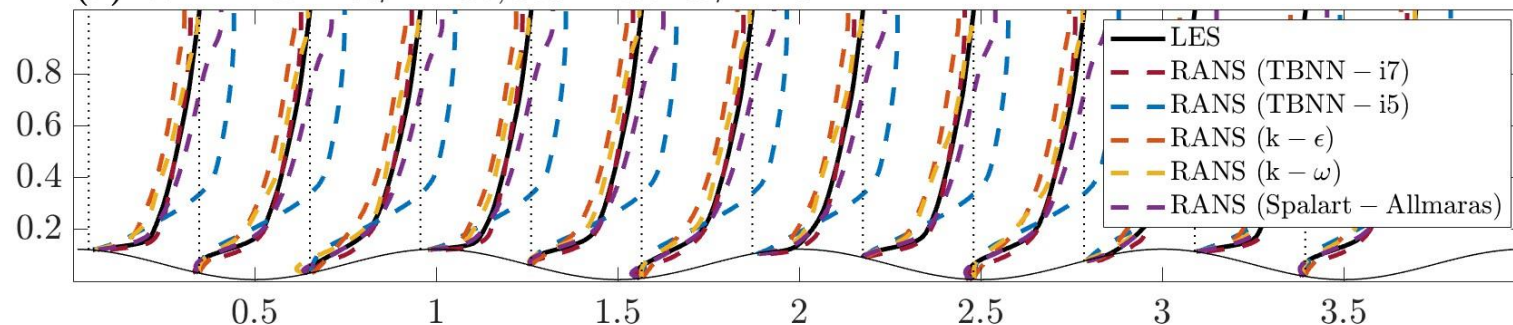


Result & Discussion - Priori part

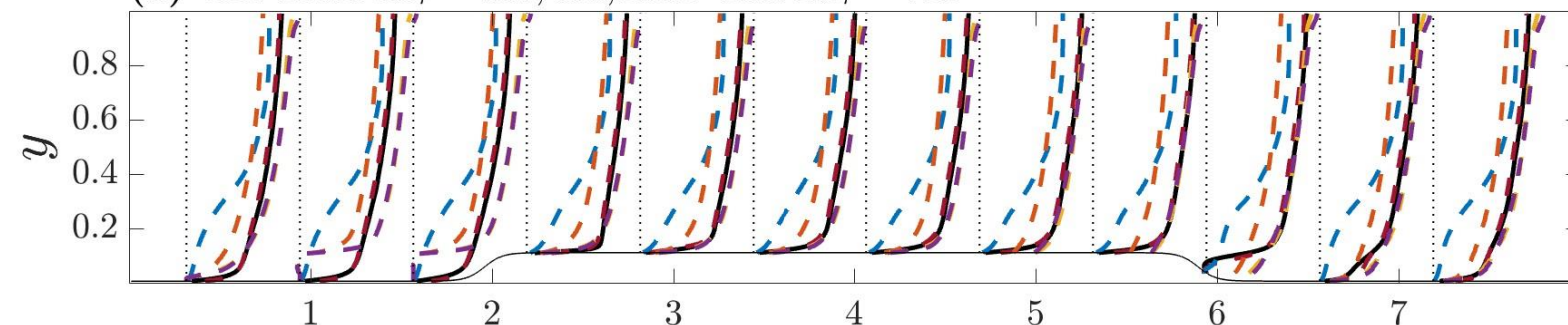


Result & Discussion - Posteriori part

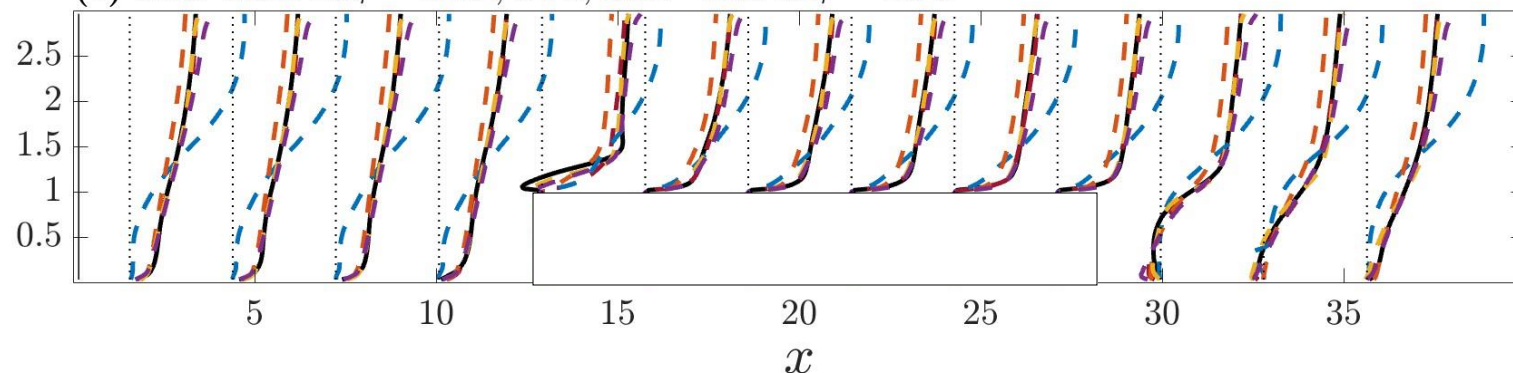
(a) WB. Train $Re_\tau = 198, 307$ Test $Re_\tau = 251$



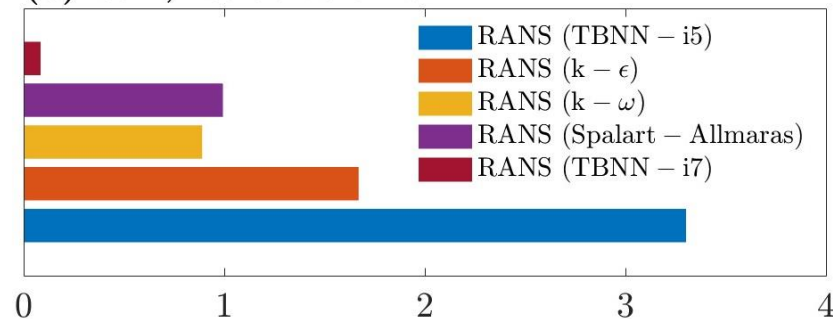
(b) CS. Train $Re_\tau = 307, 500, 1004$ Test $Re_\tau = 710$



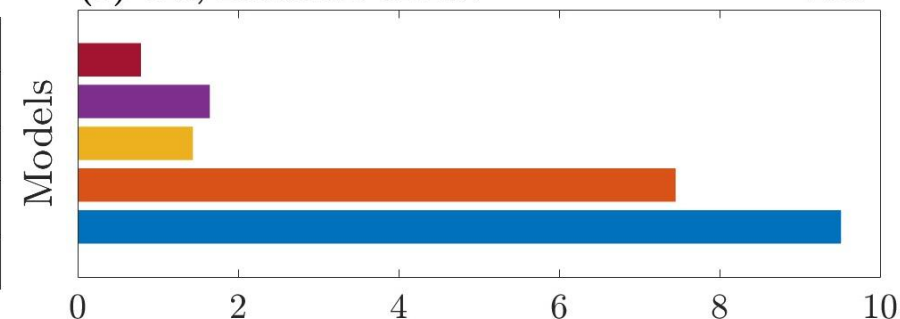
(c) RS. Train $Re_\tau = 1162, 1273, 1375$ Test $Re_\tau = 1325$



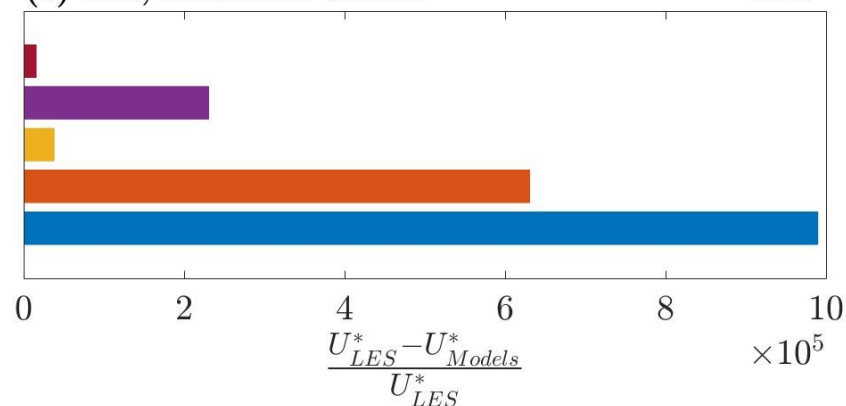
(d) WB, Relative error.



(e) CS, Relative error.



(f) RS, Relative error.



Conclusion & Future work

➤ Conclusion

- The proposed method is the first to calculate flow fields without relying on a baseline model.
- An AI model (TBNN) based on Tensor Basis Representation was developed and integrated into the RANS solver for interaction processes.
- The method was validated on three geometries: wavy bottom, curved step, and rectangular step, demonstrating high prediction accuracy.

➤ Future work

- Is it possible to shuffle all cases and perform universal prediction?
- How can we handle domain geometry in 3D cases? (e.g., when the stream function no longer exists)

References

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Thank you for your attention



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